Modelling Drug-Induced Changes in Bursting Dynamics of the *Aplysia R15* Cell

MATH3888: Projects in Mathematics (Stream 1)

Group 13

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1 Introduction

R15 is a bursting neuron in the abdominal ganglion of *Aplysia californica* (Figure 1), a sea slug found in shallow Pacific waters [1]. The neurons of the abdominal ganglion lie near its surface, with their synapses located in the central neuropil [2]. *R15* is a large caudal cell on the dorsal surface whose axon branches widely through several different nerves, consistent with a neurosecretory role in which it synthesises peptides thought to contribute to osmoregulation [2].

R15 generates robust endogenous bursting activity and consistent spiking [3, 4]. The rhythmic firing is self-generated but can also be shaped by incoming synaptic activity or circulating hormones [2], and its characteristic parabolic burst shape is altered by many transmitters and neuropeptides [3]. This makes *R15* a useful system for investigating the ionic mechanisms underlying endogenous bursting and their modulation by external signals [3]. The mathematical model considered in this study represents the principal ionic currents contributing to the electrical activity of *R15* (Figure 2).

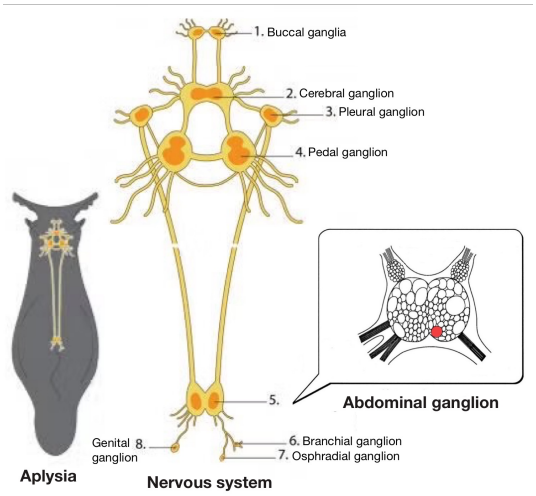


Figure 1: The nervous system and abdominal ganglion of *Aplysia*. With *R15* coloured red. Adapted from Chiu and Zare [5], Chiel Lab [6].

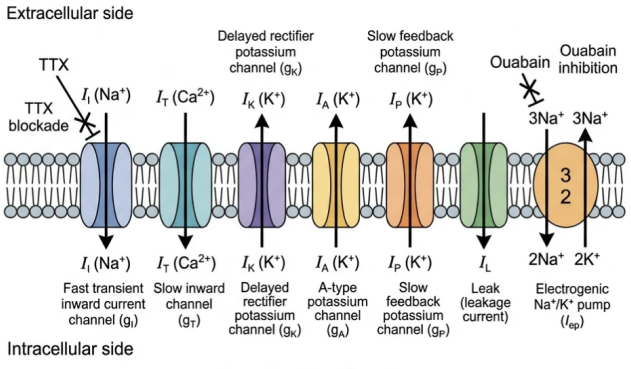


Figure 2: Close-up diagram of the ionic currents and gates in the seven-dimensional Plant–Kim [7] system.

The electrical activity of the *R15* neuron reflects the combined action of two major types of currents: inward currents that depolarise the membrane and generate spikes, and outward potassium currents that repolarise and hyperpolarise the membrane. These currents operate on different timescales. Their balance changes slowly, depolarising the membrane until the fast inward current generates a series of action potentials. The slow gate variables continue to change, gradually shifting the balance away from repetitive firing and terminating the burst. The recovery of the slow process allows the next burst to begin. [2, 8].

In this study, the Plant–Kim model [7] is used to reproduce the experimentally observed drug effects and investigate the transitions between different patterns. We analyse the effects of TTX and ouabain on the model and identify any bifurcations occur, to determine whether these bifurctaions can explain the transitions observed experimentally by Junge D [9].

2 Drugs and Experimental Data

Tetrodotoxin (TTX) and ouabain provide two pharmacological interventions to detect different components of *R15* electrical activity. TTX in calcium-free saline suppresses the transient inward current (Figure 2), which contributes to the generation of fast action potentials. In the experiment, the fast action potentials are suppressed, while the slow membrane-potential oscillations persist [9].

Ouabain inhibits the electrogenic Na^+/K^+ pump (Figure 2). In the absence of TTX, ouabain

changes normal bursting to regular firing, and artificial hyper-polarization can restore bursting [9]. In the calcium-free saline containing TTX, ouabain inhibits the remaining slow oscillations, while additional hyper-polarization current restores them [9].

These observations indicate that the transient inward current drives fast spiking, and the electrogenic pump helps to form bursting pattern. If sufficient hyper-polarising current is provided, slow oscillations can exist without pump activity Together, TTX and ouabain provide the experimental basis for the following mathematical interventions.

2.1 Experimental Protocol

The experiments of Junge D [9] were conducted on the *R15* neuron in the isolated abdominal ganglion of *Aplysia californica*. Intracellular micro-electrodes were used to record membrane potential and inject current, while experimental solutions were exchanged through continuous perfusion of the recording chamber.

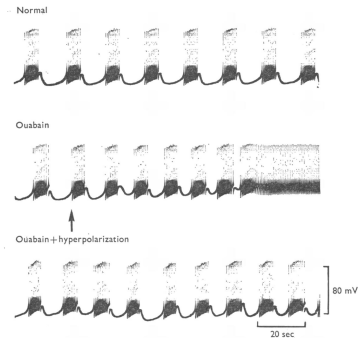
Three experimental protocols are relevant to the present study. First, membrane potential was recorded under normal conditions before applying 0.4 mM ouabain, followed by a maintained inward (hyper-polarising) current in the continued presence of ouabain (Figure 3a).

Second, normal bursting activity was compared with recordings following perfusion with calcium-free saline containing 25 μg/ml tetrodotoxin (TTX), followed by washout with normal saline to observe recovery (Figure 3b). Hyper-polarising current was also applied when necessary to maintain the slow membrane-potential waves.

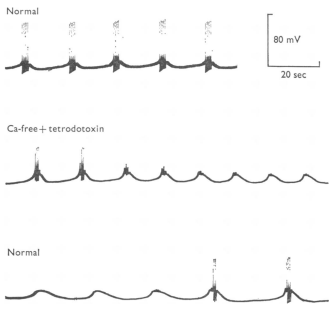
Finally, under calcium-free saline containing TTX, slow membrane-potential waves were recorded with a small maintained hyper-polarising current. Ouabain (0.4 mM) was subsequently introduced, followed by additional hyper-polarising current (Figure 3c).

2.2 Experimental Observations

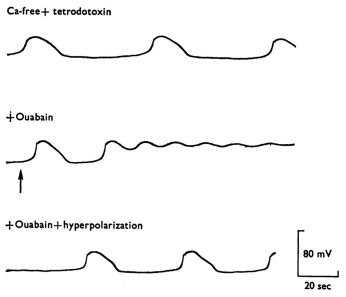
The experimental membrane-potential recordings of Junge D [9] are presented in Figure 3, illustrating the effects of TTX and ouabain individually and in combination. The experiments also include artificial hyper-polarisation, which was not explicitly investigated in our computational analysis.



(a) Ouabain (0.4 mM) alters normal bursting into regular firing. Artificial hyper-polarisation restores bursting.



(b) Calcium-free saline containing TTX (25 μg/ml) suppresses fast spikes while slow voltage waves persist. Spike activity recovers following washout.



(c) Under calcium-free saline containing TTX, addition of ouabain (0.4 mM) suppresses slow voltage waves. Additional artificial hyper-polarisation restores the waves.

Figure 3: Experimental membrane-potential recordings from the *R15* neuron of *Aplysia californica* under pharmacological interventions: (a) ouabain, (b) TTX, and (c) combined TTX and ouabain. Reproduced from Figures 2, 10, and 12, respectively, of Junge D [9].

3 Mathematical Modelling and Hypotheses

To investigate the electrical activity of *R15*, we adopt the seven-dimensional conductance-based model developed by Plant and Kim [7], which extends the classical Hodgkin–Huxley equations to describe the bursting behaviour of the neuron. The model incorporates ionic currents operating

on different timescales, allowing both fast action potentials and slower membrane-potential oscillations to be represented.

3.1 Mathematical Model

The evolution of the membrane potential V is governed by the net membrane current

$$\begin{aligned} C\dot{V} = &(g_I X_I^3 Y_I + g_T)(V_I - V) \\ &+ (g_K X_K^4 + g_A X_A Y_A + g_P X_P)(V_K - V) \\ &+ g_L(V_L - V) + I_{\text{ep}} + I_{\text{ext}}, \end{aligned} \tag{1}$$

and the six gating equations

$$\begin{aligned} \dot{X}_I &= \frac{S_I(V) - X_I}{\tau_{X_I}(V)}, \qquad \dot{Y}_I = \frac{Z_I(V) - Y_I}{\tau_{Y_I}(V)}, \qquad \dot{X}_K = \frac{S_K(V) - X_K}{\tau_{X_K}(V)}, \\ \dot{X}_A &= \frac{S_A(V) - X_A}{\tau_{X_A}}, \qquad \dot{Y}_A = \frac{Z_A(V) - Y_A}{\tau_{Y_A}}, \qquad \dot{X}_P = \frac{S_P(V) - X_P}{\tau_{X_P}}. \end{aligned} \tag{2}$$

Here, C denotes the membrane capacitance, while V_I, V_K, and V_L are the reversal potentials for the inward, potassium, and leak currents, respectively. The conductance parameters g_i determine the contributions of the corresponding ionic currents to the membrane voltage.

The inward-current term in Equation (1) consists of a transient component $g_I X_I^3 Y_I$, responsible for fast action-potential generation, and a persistent component g_T that remains active under TTX blockade. The outward potassium currents comprise the fast potassium conductance $g_K X_K^4$, the inactivating A-type conductance $g_A X_A Y_A$, and the very slow conductance $g_P X_P$. In particular, X_P evolves on a substantially longer timescale than the other gating variables, providing slow feedback that contributes to the underlying membrane-potential oscillations.

The remaining terms represent the passive leak current, the hyper-polarising electrogenic pump current I_{ep}, and the externally applied current I_{ext}. The gating variables X_i and Y_i describe channel activation and inactivation, relaxing towards voltage-dependent steady-state functions $S_i(V)$ and $Z_i(V)$ with time constants τ_{X_i} and τ_{Y_i}. All gating functions, time constants, conductances and reversal potentials are taken unchanged from Plant and Kim [7], except for g_I and I_{ep}, which are varied as described in Section 3.2 (untreated values $g_I^{(0)} = 4.0 \times 10^{-3}$ mho and $I_{\text{ep}}^{(0)} = -0.22\,\mu$A). The slow gate X_P is given explicitly in Equation (3). Throughout our computational drug-intervention analyses, $I_{\text{ext}} = 0$.

3.2 Drug Parametrisation and Mathematical Hypotheses

The effects of TTX and ouabain are represented by scaling the transient inward conductance and electrogenic pump current, respectively as $\boldsymbol{g_I = \alpha_I g_I^{(0)}}$ and $\boldsymbol{I_{\text{ep}} = \alpha_P I_{\text{ep}}^{(0)}}$, where the untreated parameter values are $g_I^{(0)} = 4.0 \times 10^{-3}$ mho and $I_{\text{ep}}^{(0)} = -0.22\,\mu$A.

The scaling parameters α_I and α_P represent the fractions of the corresponding model conductance and pump-current magnitude remaining. For either parameter, a value of 1 denotes the untreated condition, values between 0 and 1 represent partial inhibition, and 0 denotes complete blockade. Thus, the biologically relevant intervention region is $(\alpha_I, \alpha_P) \in [0,1]^2$. These parameters describe the extent of inhibition within the model rather than experimentally measured drug concentrations. Continuation beyond this region is also considered to characterise the wider mathematical bifurcation structure.

Hypothesis 1: TTX. With $\alpha_P = 1$ fixed, decreasing α_I towards zero reduces the transient inward conductance. We hypothesise that this suppresses fast action potentials while allowing the underlying slow membrane-potential oscillations to persist. Equilibrium continuation in α_I is used to investigate whether bifurcations accompany these changes in the simulated dynamics.

Hypothesis 2: Ouabain. With $\alpha_I = 1$ fixed, decreasing α_P towards zero reduces the magnitude of the hyper-polarising electrogenic pump current. We hypothesise that this alters the normal bursting pattern and promotes a transition towards regular beating. Continuation in α_P is used to examine the associated equilibrium structure and possible bifurcations.

Hypothesis 3: Combined TTX and ouabain. Under complete TTX blockade ($\alpha_I = 0$), further reduction of α_P is expected to suppress the remaining slow oscillations and eventually produce a stable equilibrium. We investigate this through equilibrium and periodic-orbit continuation in α_P.

4 Computational Modelling Results

4.1 Baseline Dynamics

The untreated Plant–Kim model was integrated numerically in `Python` using the `VODE` integrator, with the resulting trajectories visualised using `Matplotlib`. Figure 4 shows the simulated membrane voltage $V(t)$ under control conditions. Following an initial transient burst, the model exhibits sustained, approximately periodic bursting, characterised by brief trains of fast action potentials separated by longer hyper-polarised silent intervals. The mean bursting period, active-phase duration, and silent-phase duration were computed from the post-transient trajectory and are indicated in the figure.

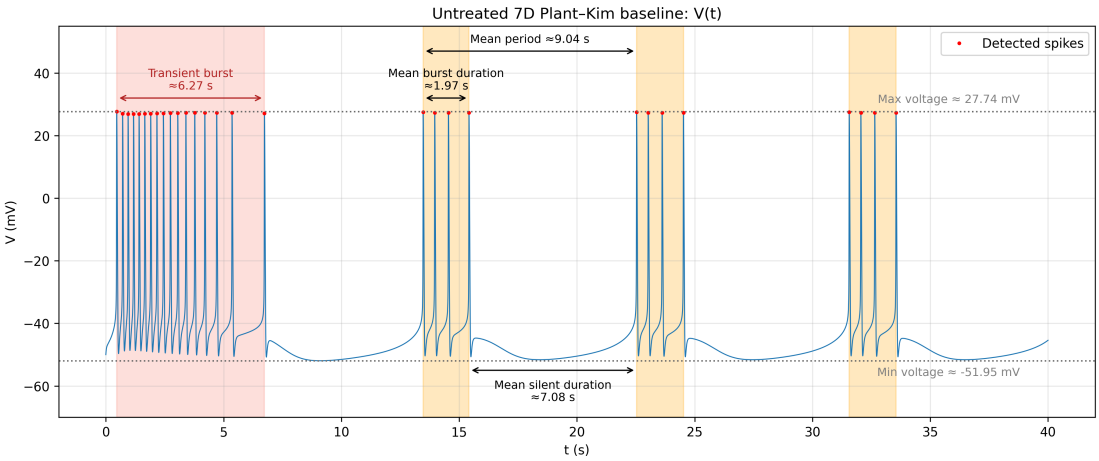


Figure 4: Simulated membrane voltage $V(t)$ of the untreated seven-dimensional Plant–Kim model. The initial transient burst is highlighted in red, while subsequent active phases are shaded yellow. Red markers indicate detected spikes, and arrows show the mean bursting period, active-phase duration, and silent-phase duration computed from the post-transient response. Dotted horizontal lines indicate the observed voltage extrema.

To examine the relationship between membrane voltage and channel gating, we consider six phase-plane projections of the post-transient trajectory, shown in Figure 5. The activation variables X_K, X_A, and X_I, together with the inactivation variables Y_A and Y_I, exhibit distinct looping trajectories reflecting their coupled evolution with membrane voltage and differing gating timescales. In particular, Y_A exhibits an intermediate response timescale relative to the faster activation variables.

The very slow potassium activation variable X_P, exhibits nearly vertical voltage excursions in its phase-plane projection. V changes rapidly while X_P barely moves during a spike, although it evolves gradually over the longer bursting cycle. Together with its long time constant, this motivates a fast–slow decomposition in which X_P is treated as a quasi-static parameter of the remaining six-dimensional subsystem.

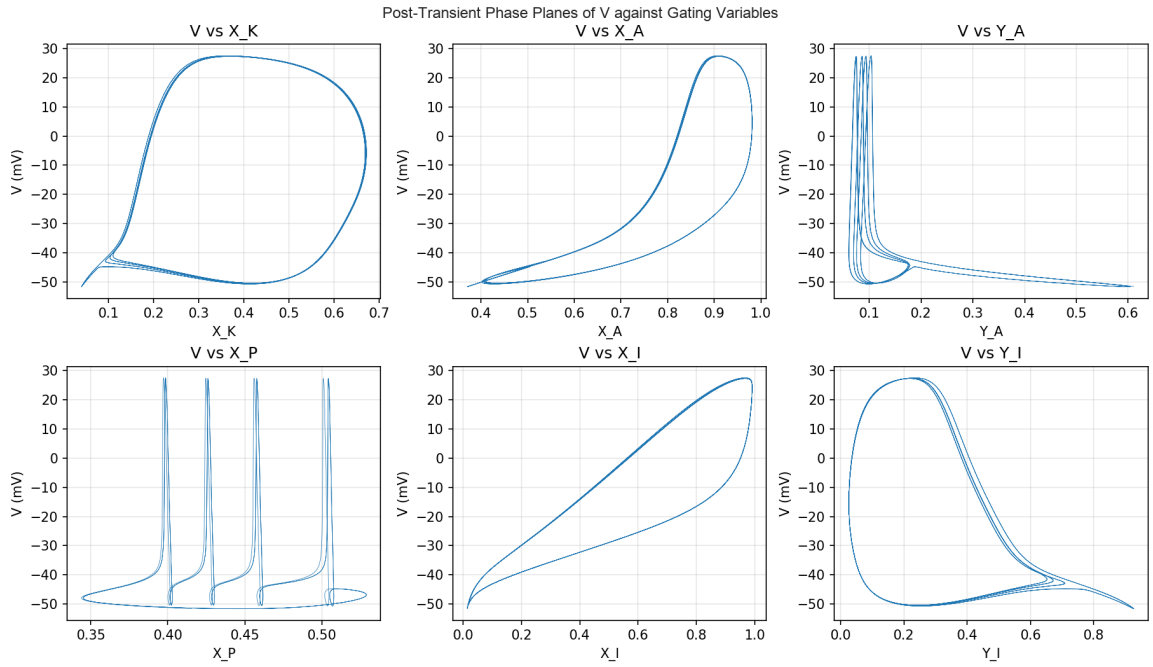


Figure 5: Post-transient phase-plane projections of membrane voltage V against the six gating variables X_K, X_A, Y_A, X_P, X_I, and Y_I. Each panel represents a two-dimensional projection of the same seven-dimensional simulated trajectory. The differing loop geometries illustrate the coupled voltage–gating dynamics, with the V–X_P projection particularly highlighting the slow evolution of X_P relative to rapid voltage excursions.

4.2 Fast–Slow Decomposition

The slow potassium activation variable X_P obeys

$$\frac{\mathrm{d}X_P}{\mathrm{d}t} = \frac{S_P(V) - X_P}{\tau_{X_P}}, \qquad S_P(V) = \frac{1}{1 + e^{-0.7(V+47)}}, \qquad \tau_{X_P} = 8000\,\mathrm{ms}. \tag{3}$$

Its time constant is substantially larger than those of the other gating variables, with the next slowest being $\tau_{Y_A} = 235\,\mathrm{ms}$. During the active phase, X_P changes at approximately $7 \times 10^{-5}\,\mathrm{ms}^{-1}$, corresponding to a change of less than 10^{-3} over a typical 5 ms action potential. Thus, X_P can be treated as approximately constant on the timescale of individual spikes.

By fixing X_P as a parameter and removing its differential equation, we obtain a six-dimensional fast subsystem in $(V, X_K, X_A, Y_A, X_I, Y_I)$. Examining how its equilibria and periodic solutions vary with X_P provides a framework for understanding the bursting mechanism, following the approach of Rinzel [10] and Canavier et al. [1].

Since X_P enters the voltage equation linearly through $g_P X_P (V_K - V)$, the equilibria can be obtained by setting the remaining gating variables to their steady-state values and solving the voltage balance algebraically for X_P as a function of V. Sweeping V traces the equilibrium curve, including its unstable branches, while stability is determined from the eigenvalues of the six-dimensional Jacobian.

At control, Figure 6a shows that the equilibrium curve is Z-shaped, with a fold at $(X_P, V) \approx (0.4003, -49.63\,\mathrm{mV})$. Above this fold, three equilibria coexist: a stable resting state, a saddle, and an unstable depolarised equilibrium. At the fold, the resting state and saddle collide and disappear. A subcritical Hopf bifurcation also occurs on the depolarised branch at $(X_P, V) \approx (0.0265, -14.59\,\mathrm{mV})$.

Periodic solutions were identified by simulating the fast subsystem at successive fixed values of X_P, using the preceding simulation endpoint as the next initial condition. Stable spiking solutions were

observed up to $X_P \in [0.5025, 0.5050]$. Approaching this endpoint, the interspike interval increased from 1130 ms at $X_P = 0.4875$ to 2458 ms at $X_P = 0.5025$, while the spike amplitude remained approximately constant. Fitting the logarithmic scaling law $\boldsymbol{T = a\,[-\ln(X_P^c - X_P)] + b}$ gave $X_P^c \approx 0.5036$ and $a \approx 490$ ms, with correlation 0.99995. The inverse fitted coefficient, $1/a \approx 2.04 \times 10^{-3}\,\mathrm{ms}^{-1}$, is comparable to the unstable saddle eigenvalue $1.67 \times 10^{-3}\,\mathrm{ms}^{-1}$. These findings are consistent with a homoclinic termination of the spiking branch.

Between the resting-state fold and the estimated spiking termination, approximately $0.4003 < X_P < 0.5036$, the fast subsystem exhibits bistability between resting and spiking solutions. This coexistence was also supported by simulations with initial conditions on opposite sides of the saddle.

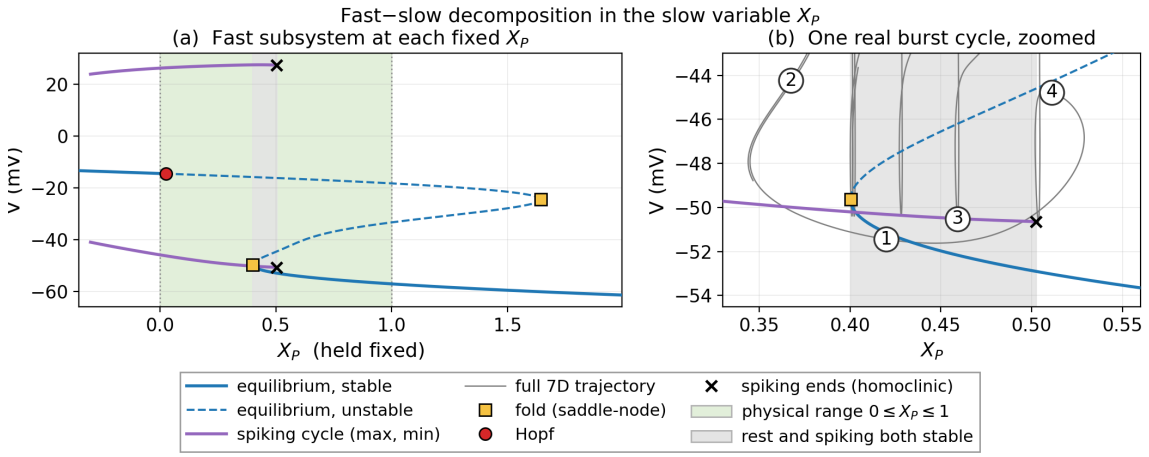


Figure 6: Fast–slow decomposition of the untreated Plant–Kim model. (a) Bifurcation diagram of the six-dimensional fast subsystem with X_P fixed. (b) One bursting cycle of the full seven-dimensional model projected onto the same (X_P, V) coordinates, with numbered stages indicating (1) silence, (2) burst onset, (3) active spiking, and (4) burst termination.

The full seven-dimensional bursting trajectory is superimposed on the fast-subsystem structure in Figure 6b. During the silent phase, the membrane potential remains near −52 mV, below the half-activation voltage of −47 mV. Consequently, $S_P(V) < X_P$ and X_P gradually decreases. As the trajectory passes the resting-state fold, it eventually enters the active phase, where sustained spiking increases X_P through slow potassium activation. The active phase then terminates as X_P approaches the estimated spiking-branch endpoint.

The fast-subsystem boundaries were obtained independently of the full bursting simulation. Nevertheless, the post-transient trajectory visits $X_P \in [0.344, 0.529]$, with its final spike occurring at $X_P \approx 0.502$, close to the predicted spiking termination near 0.5036. The trajectory also overshoots the resting-state fold, reaching $X_P \approx 0.345$ before transitioning into the active phase. This delayed transition is consistent with the finite separation of timescales, under which the full trajectory need not switch precisely at the frozen-subsystem bifurcation point.

4.3 TTX Intervention Results

TTX inhibition was modelled by decreasing $\alpha_I = g_I / g_I^{(0)}$ from 1 to 0 with $\alpha_P = 1$ fixed, where $g_I^{(0)} = 4$ (mho $\times 10^{-3}$). The intervention range $\alpha_I \in [0, 1]$ corresponds to $g_I \in [0, 4]$; values outside it are used only to show the full structure. We expect fast spiking to disappear while the slow oscillation remains [9].

4.3.1 Time Traces

Figure 7 shows the simulated membrane voltage $V(t)$ for selected values of α_I. Under control ($\alpha_I = 1$) the model bursts regularly with a period of approximately 9 s. As α_I decreases, the number and amplitude of spikes in each burst diminish, with the peak voltage in the later bursts

falling from approximately 28 mV at $\alpha_I = 1$ to -2 mV at $\alpha_I = 0.25$, while a slow oscillation between bursts remains.

At $\alpha_I = 0.2$, fast spikes occur only during the initial transient, after which a slow oscillation persists without further action potentials. This oscillation remains under complete TTX blockade ($\alpha_I = 0$), with peaks roughly 10 s apart.

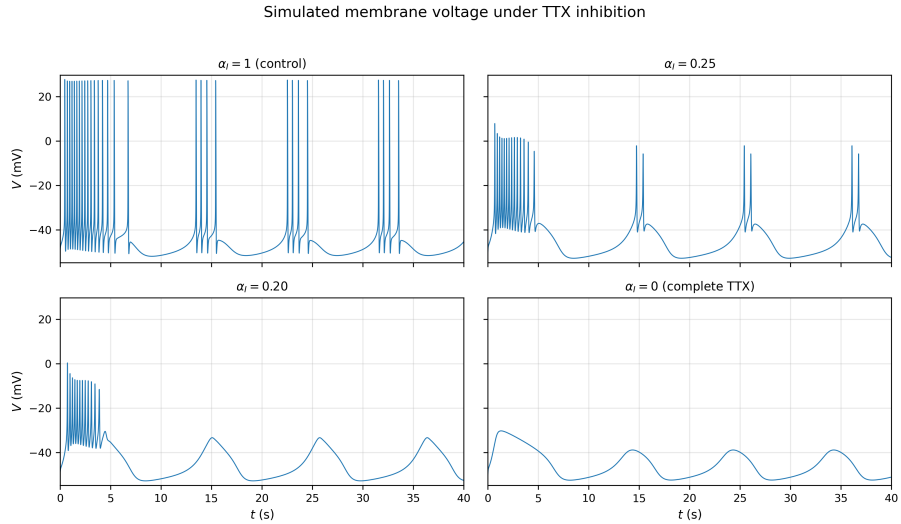


Figure 7: Membrane voltage $V(t)$ under TTX inhibition, $\alpha_I = g_I/g_I^{(0)}$ ($\alpha_P = 1$, $I_{\text{ext}} = 0$, identical initial conditions). Spikes shrink and then vanish as α_I decreases, leaving only the slow oscillation.

4.3.2 Preliminary Analysis

Method: Equilibria of the full seven-dimensional model were followed by pseudo-arclength continuation (`PyDSTool` EP-C) from the untreated equilibrium at $V^* = -18.312$ mV, forwards and backwards over $g_I \in [-60, 220]$ ($\alpha_I \in [-15, 55]$). Limit cycles were continued from each Hopf point with `AUTO` (LC-C). Hopf points were classified by the sign of the first Lyapunov coefficient ℓ_1 ($\ell_1 < 0$ supercritical, $\ell_1 > 0$ subcritical), computed by finite differences.

Equilibria: At control ($\alpha_I = 1$) the model has three unstable equilibria ($V^* = -47.4$, -33.3 and -18.3 mV) on a single S-shaped branch with two folds and two Hopf points (Figure 8).

The folds split the branch into a hyper-polarised lower branch (existing for $\alpha_I < 12.9$), a middle branch of saddles, and a depolarised upper branch (existing for $\alpha_I > 0.614$). Only the upper branch above H_{pos} and the lower branch below H_{neg} are stable: both lie outside $[0, 1]$ and trajectories settle on them.

Within $\alpha_I \in [0, 1]$ every equilibrium is unstable, there is no Hopf point, and the only bifurcation is LP_{bio} at $\alpha_I = 0.614$, where the upper and middle equilibria merge and disappear. Between the two Hopf points no resting state is stable, so the attractor must be non-stationary: the slow wave and bursting seen in the traces.

Limit cycles: A limit cycle is born at each Hopf point (Figure 8). At H_{neg} it is stable near the Hopf point and grows in amplitude as α_I increases, with its period rising from about 7.2 s to about 11.8 s where continuation was stopped ($\alpha_I \approx 0.31$), comparable to the 9-10 s slow-wave period in Figure 7. At H_{pos} it is unstable, with a period of about 78 ms that rises steeply to about 79 s by $\alpha_I \approx 3.36$, where continuation was stopped. Crosses in Figure 8 mark where each branch was stopped, and both are reliable only near their Hopf points.

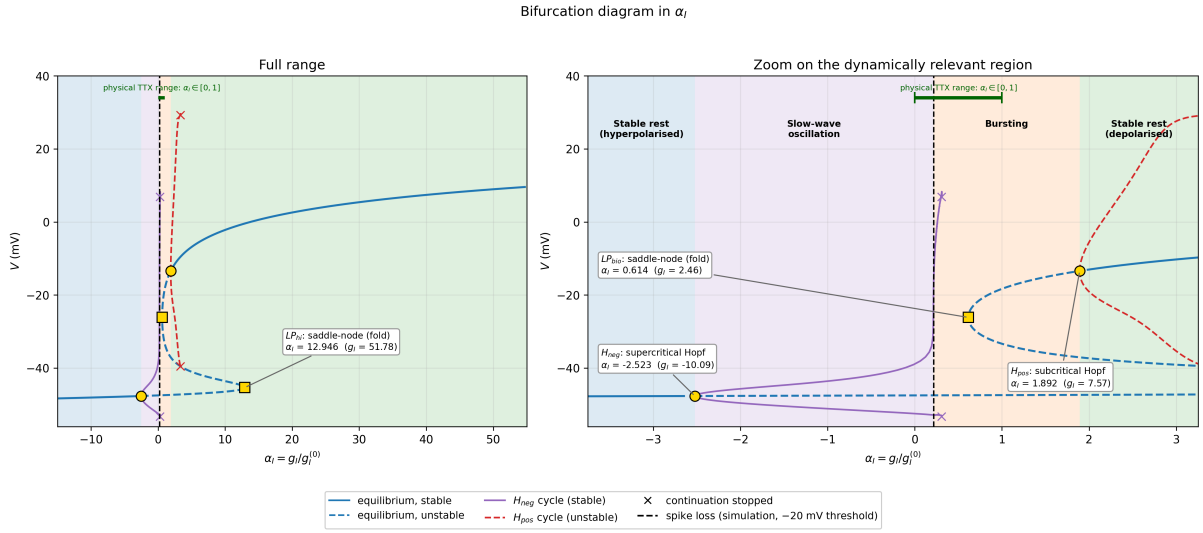


Figure 8: Bifurcation diagram in $\alpha_I = g_I/g_I^{(0)}$ ($g_I^{(0)} = 4\,\text{mho} \times 10^{-3}$) with $\alpha_P = 1$ fixed. Shaded regions show the regimes observed in simulation. Left: full range; right: zoom.

Spike-disappearance threshold: Equilibrium continuation shows no change between $\alpha_I = 0.2$ and 0.25, even though the traces show spiking disappearing there, so we located the threshold by simulation. After discarding a 20 s transient, we counted upward crossings of -20 mV over the next 30 s and used bisection, giving $g_I \approx 0.861$ ($\alpha_I \approx 0.215$, dashed black line in Figure 8). Below this only the slow oscillation remains, and above it the model also bursts. The loss of spiking is gradual, with the late-run peak voltage rising smoothly from about -33 mV at $\alpha_I = 0.2$ to -2 mV at 0.25, so $\alpha_I \approx 0.215$ is an approximate threshold that depends on the chosen -20 mV level, not a sharp transition.

4.3.3 Relation to the Slow Variable

Repeating the decomposition at complete blockade ($\alpha_I = 0$), the equilibrium curve keeps its S shape, with folds at $X_P = 0.396$ and $X_P = 0.499$, but for all X_P the stable spiking limit cycle vanishes (Figure 9d). The slow variable now moves in a cycle between the two stable branches of the curve so what's left is a relaxation oscillation of the slow subsystem and not a series of action potentials. The full trajectory visits $X_P \in [0.340, 0.576]$, a little wider than the untreated range.

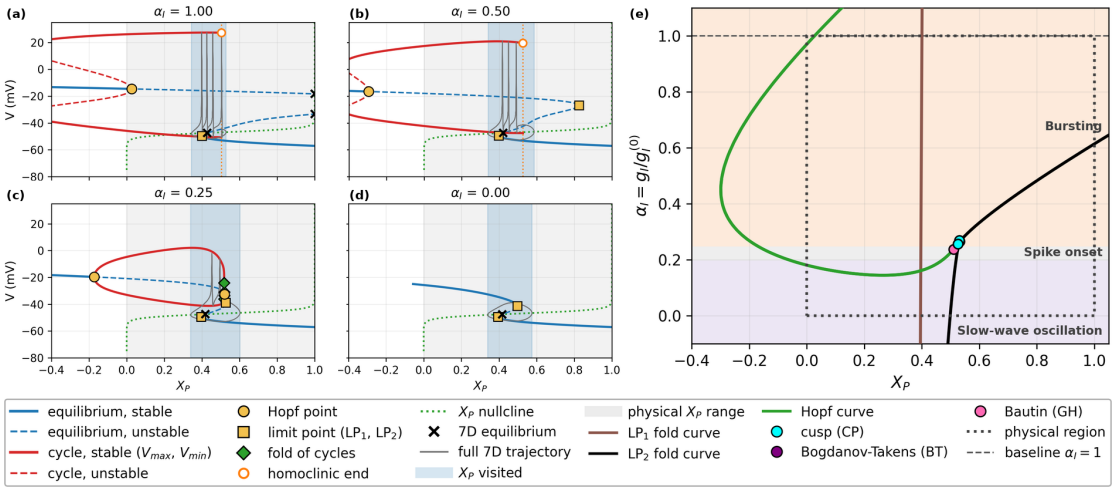


Figure 9: (a)–(d) Fast-subsystem bifurcation diagrams in X_P for $\alpha_I = 1, 0.5, 0.25$ and 0, each a horizontal slice of the two-parameter diagram in (e). (e) Fold curves LP_1 and LP_2, Hopf curve and codimension-two points in (X_P, α_I), with the dynamical regimes from 7D simulation shaded. The dotted rectangle marks the physical region.

The loss of fast spiking is thus caused by the destruction of the stable spiking branches. The continuation in α_I of Section 4.3.2 cannot see this, as it follows equilibria while the object that disappears is a periodic orbit. This explains why $\alpha_I \approx 0.215$ had to be found by simulation.

To further understand the transition to oscillations observed between $0.2 < \alpha_I < 0.25$, the equilibrium curve for decreasing α_I is observed in Figure 9a–d which is a horizontal slice of the two parameter diagram in Figure 9e.

Firstly, the upper fold (LP_2) moves to smaller X_P, approaching the lower fold (LP_1, which remains around $X_P \approx 0.4$) at a steady rate until $\alpha_I \approx 0.25$, then more slowly until the two merge at a cusp point when $\alpha_I = -4.6075$ and $X_P = 0.367$. Once LP_2 lies inside the range of X_P visited by the 7D trajectory, the trajectory is turned around by the two folds before it can reach the limit cycle. This relaxation oscillation needs only the folds, not spiking, so it persists and supplies the slow wave even after spikes are lost.

Secondly, the spiking amplitude decreases as the Hopf bubble shrinks and changes from subcritical to supercritical at the Bautin point ($\alpha_I \approx 0.24$). Below this, a second Hopf point (H_2) emerges from LP_2 at a Bogdanov-Takens point ($\alpha_I \approx 0.26$), and the cycle branch from H_1 now ends at H_2 rather than at a homoclinic orbit. As α_I decreases, the two Hopf points merge and disappear leaving no spikes and only the oscillation between the two limit points.

4.4 Ouabain Intervention Results

Ouabain inhibition was modelled by decreasing α_P from 1 (untreated control) to 0 (complete pump blockade), while fixing $\alpha_I = 1$. This reduces the hyper-polarising electrogenic pump current. We examined the resulting membrane-voltage dynamics and bifurcation structure through numerical simulations and continuation.

4.4.1 Time Traces

Figure 10 shows 60 s voltage traces from identical initial conditions. At $\alpha_P = 1$, the model exhibits regular bursting, with approximately four spikes per burst and a burst period of 9 s.

At $\alpha_P = 0.5$, initial continuous spiking develops into bursts with varying spike numbers and spacing. At $\alpha_P = 0.2$, the initial spiking phase lasts longer before the spikes become sparse and irregular. Under complete blockade ($\alpha_P = 0$), clear inter-burst silent intervals are absent, and spikes recur approximately every 0.5 s during the final 20 s, producing nearly regular continuous firing.

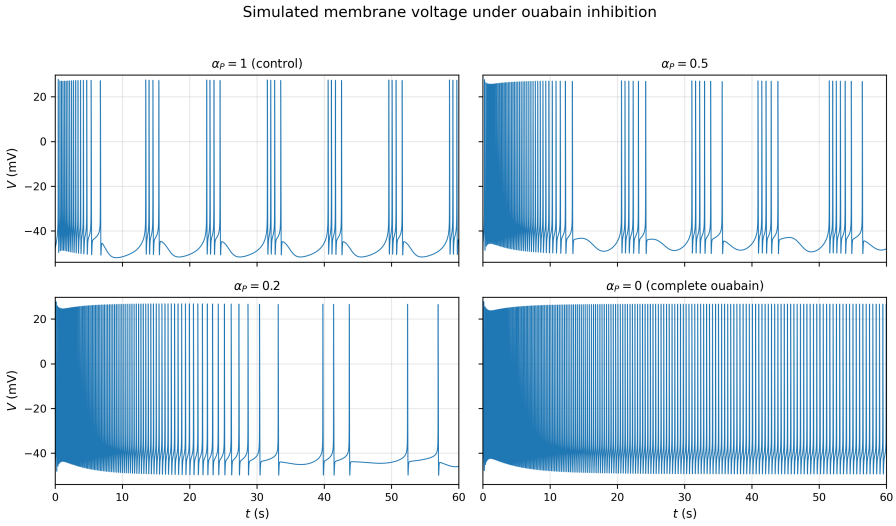


Figure 10: Membrane-voltage traces for decreasing $\alpha_P = I_{ep}/I_{ep}^{(0)}$, where $I_{ep}^{(0)} = -0.22\,\mu$A. All simulations use identical initial conditions and a duration of 60 s.

4.4.2 Preliminary Analysis

Method. Equilibria were continued forwards and backwards in I_{ep} using `PyDSTool EP-C`, starting from the control equilibrium at $V^* \approx -18.312\,\mathrm{mV}$, with all other parameters fixed. Results are plotted against $\alpha_P = I_{ep}/I_{ep}^{(0)}$. Stability was assessed using Jacobian eigenvalues, while first Lyapunov coefficients, calculated from analytical derivatives, determined whether each Hopf bifurcation was supercritical or subcritical. Periodic-orbit branches were continued from both Hopf points using `LC-C` with `AUTO`.

Equilibria. Numerical continuation identifies an S-shaped equilibrium curve with two folds and two Hopf bifurcations (Figure 11). The folds, LP_{bio} at $\alpha_P \approx 0.0941$ and LP_{ext} at $\alpha_P \approx 3.2234$, divide the curve into lower-, middle- and upper-voltage branches.

Within the computed range, the lower branch is unstable between LP_{bio} and H_{ext2}, and asymptotically stable for $\alpha_P > 1.5041$. The middle branch connects the two folds and is unstable throughout. The upper branch is unstable between LP_{ext} and H_{ext}, and asymptotically stable for $\alpha_P < -2.0125$. The Hopf bifurcations at H_{ext2} and H_{ext} are supercritical and subcritical, respectively.

Three equilibria coexist at control ($\alpha_P = 1$), and all equilibria within the biological interval $\alpha_P \in [0, 1]$ are unstable. As pump inhibition increases and α_P decreases, the lower and middle equilibria coalesce at LP_{bio} and disappear, leaving a single unstable upper equilibrium.

Limit cycles. Continuation from the two Hopf points (Figure 11) reveals two periodic-orbit families with distinct timescales. The cycle from H_{ext} is unstable near the subcritical Hopf point. Its amplitude grows as α_P decreases, reaching a fold of limit cycles at $\alpha_P \approx -2.946$, where the period is 114.9 ms. The branch then turns towards increasing α_P. Its period rises from 94.2 ms near Hopf to 129.9 ms at the numerical endpoint.

The cycle from H_{ext2} is stable near the supercritical Hopf point. Its amplitude and period grow as α_P decreases, with the period rising from approximately 11.7 s near Hopf to 56 s at the numerical endpoint ($\alpha_P \approx 1.454$).

Both computed families lie outside the biological interval. Stability has been established only locally near the Hopf points; stability elsewhere along the branches has not been verified. The numerical endpoints mark where continuation was stopped, rather than confirmed termination of the periodic orbits.

Within the biological interval, all computed equilibria are unstable, while the voltage traces show bursting at $\alpha_P = 1$ and nearly regular continuous spiking at $\alpha_P = 0$. The fold at $\alpha_P \approx 0.0941$ changes the number of equilibria but does not by itself locate the transition between these firing patterns.

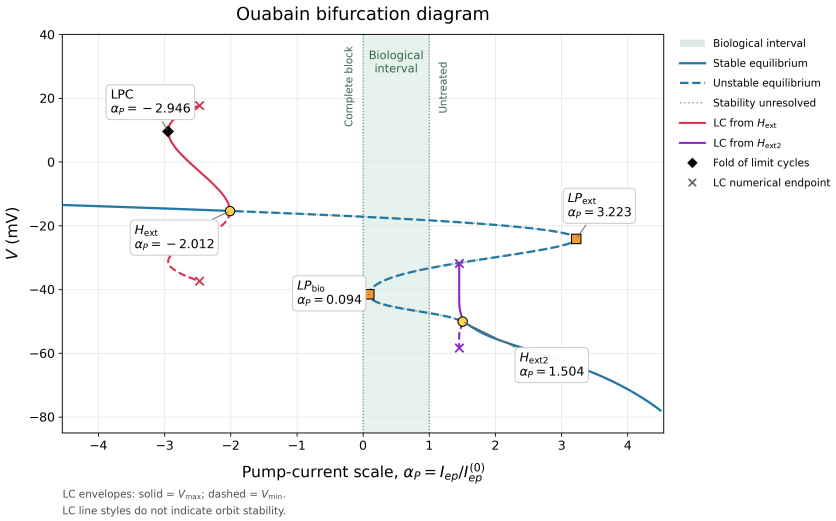


Figure 11: Equilibrium and periodic-orbit branches under pump-current variation, plotted against $\alpha_P = I_{ep}/I_{ep}^{(0)}$, where $I_{ep}^{(0)} = -0.22\,\mu$A. The shaded region marks the biological interval $\alpha_P \in [0, 1]$. Periodic-orbit envelopes show the maximum and minimum membrane voltages along each orbit. The computed fold of limit cycles occurs at $\alpha_P \approx -2.946$.

4.4.3 Relation to the Slow Variable

At complete pump blockade ($\alpha_P = 0$) the fast subsystem has a stable spiking solution and a rest state still remains but the whole structure has shifted to larger X_P (Figure 12). The fold moves from $X_P = 0.400$ to $X_P = 0.960$, and the Hopf on the depolarised branch from $X_P = 0.027$ to $X_P = 0.351$. The rest state therefore exists only above $X_P = 0.960$, while the full trajectory visits only $X_P \in [0.783, 0.897]$.

For the cell to enter its silent phase it must reach the rest branch. Here it is not possible; the rest state exists, but cannot at any X_P reach it, so nothing ends the active phase and firing continues indefinitely. This accounts for the change from bursting to continuous beating that the continuation in α_P of Section 4.4.2 couldn't attribute to any bifurcations with $\alpha_P \in [0, 1]$.

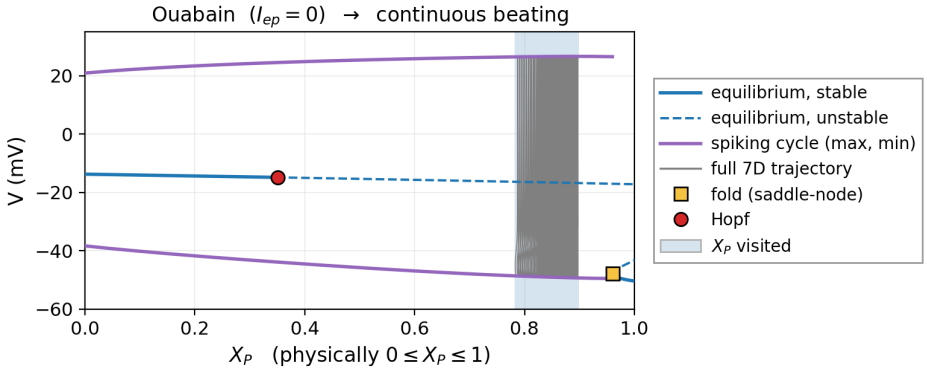


Figure 12: Fast-subsystem structure in X_P under complete ouabain blockade ($\alpha_P = 0$). The fold has moved to $X_P = 0.960$ and the Hopf to $X_P = 0.351$, carrying the rest branch beyond the range of X_P the full trajectory reaches, so the active phase has no point and firing does not stop.

To examine how this change develops under partial pump inhibition, Figure 13 shows four fixed-α_P slices of the two-parameter diagram alongside the full-model trajectories.

Firstly, the lower fold moves towards larger X_P, from approximately 0.685 at $\alpha_P = 0.5$ to 0.911 at $\alpha_P = 0.09$. The trajectory also changes: grouped spikes span a relatively broad X_P range at $\alpha_P = 0.5$, whereas repeated spikes occupy much narrower ranges at $\alpha_P = 0.1$ and 0.09. Under

complete blockade, the fold lies beyond the visited range, placing the lower stable resting branch outside the trajectory's excursions, as discussed above.

Secondly, the Hopf point also shifts towards larger X_P, from approximately 0.189 at $\alpha_P = 0.5$ to 0.322 at $\alpha_P = 0.09$. However, it remains well below the visited ranges. The transition therefore does not appear to involve the trajectory crossing this Hopf point. Instead, the diagrams support an interpretation in terms of the changing relationship between the trajectory and the lower resting branch.

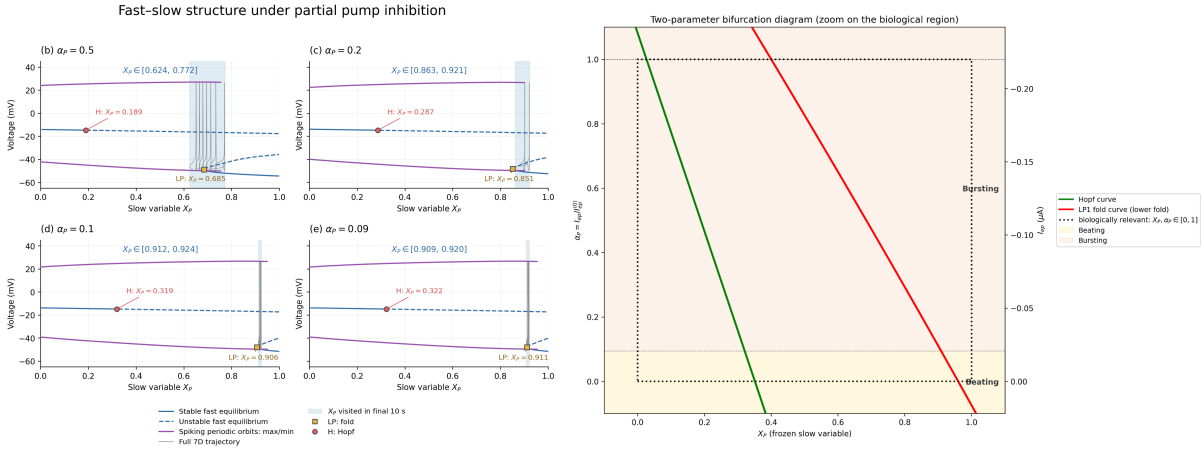


Figure 13: Fast–slow dynamics under pump inhibition. Left: fast-subsystem diagrams at $\alpha_P = 0.5$, 0.2, 0.1 and 0.09. Labels indicate the X_P coordinates of Hopf points (H) and folds (LP). Grey curves show full-model trajectories over the final 10 s, with shaded regions marking the visited X_P ranges. Right: Hopf and lower-fold curves in the (X_P, α_P) plane; background shading indicates firing regimes from full-model simulations.

4.5 Combined Drug Intervention Results

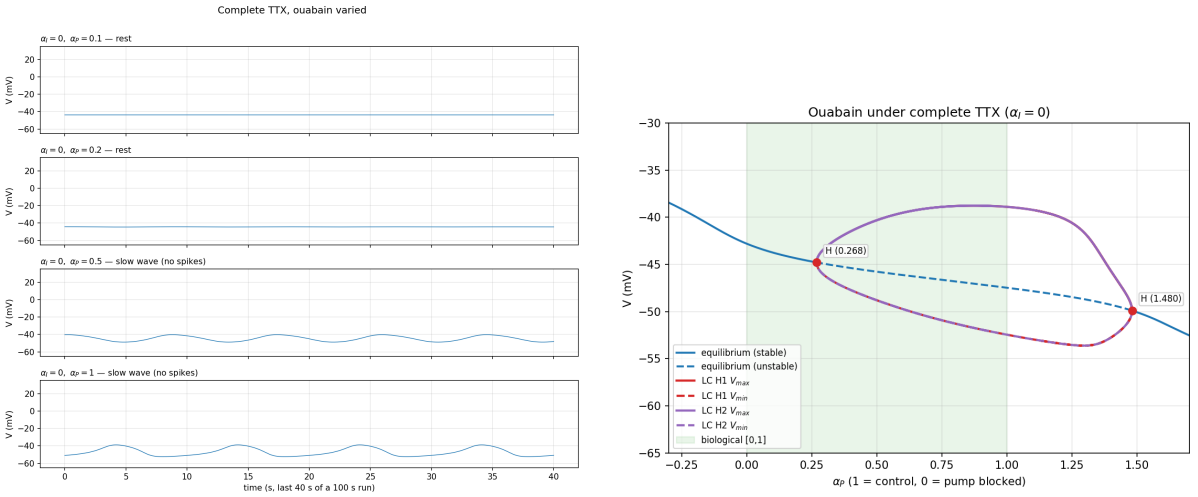
In order to test hypothesis 3 and replicate the experiment in Figure 3c, we fix complete TTX blockade ($\alpha_I = 0$) and reduce the pump parameter α_P from 1 to 0.

4.5.1 Time Traces

The simulation of the seven-dimensional model using `VODE` began from $V_0 = -50$mV and ran for 100s of which only the final 40 s is shown. Figure 14a shows that under complete TTX ($\alpha_I = 0$) we observe a non-spiking slow wave regime from $\alpha_P = 1$ to 0.5, with peak to peak amplitude ranging from 13.6 to 8.5 mV while over the same region the period remained around 9–10 s. By $\alpha_P = 0.2$ the oscillation has ceased and the voltage comes to rest near -44 mV, this agrees with the behaviour recorded by Junge D [9] upon adding ouabain.

4.5.2 Preliminary Analysis

The continuation in α_P (EP-C then LC-C from the Hopf points, as discussed in Sections 4.3.2 and 4.4.2) reveals a single equilibrium, thus the stability is determined by the presence of Hopf points (Figure 14b). At $\alpha_P = 1$ it is unstable ($V^* = -47.48$ mV), hence the voltage is not able to settle and so circles the equilibrium periodically, and so we see the slow wave in the time trace. There are no spikes in the orbit because, through application of TTX, the transient inward current is completely blocked. Then as α_P is decreased we observe the orbit shrink to zero amplitude at a Hopf point labelled H_{bio} ($\alpha_P = 0.268$, $V = -44.79$ mV), beyond H_{bio} the equilibrium is stable, corresponding to the time traces showing a steady rest potential. Outside the biologically meaningful region there is a second Hopf point labelled H_{ext} ($\alpha_P = 1.480$) which closes the orbit into a Hopf bubble. So under complete TTX inhibition of the electrogenic pump behaves like a switch with slow wave behaviour ended by a Hopf bifurcation.



(a) $V(t)$ for $\alpha_P = 0.1, 0.2, 0.5, 1$.

(b) Continuation in α_P.

Figure 14: Varying ouabain under complete TTX. Figure (a) shows the final 40s of the voltage traces. Figure (b) shows the Hopf points and periodic orbit between H_{bio} ($\alpha_P = 0.268$) and H_{ext} ($\alpha_P = 1.480$).

4.5.3 Relation to the Slow Variable

At combined blockade ($\alpha_I = \alpha_P = 0$) no spiking solution exists at any X_P, and the equilibrium curve keeps only a very narrow S-bend between $X_P = 0.950$ and $X_P = 0.955$ (Figure 15). The trajectory settles at $X_P = 0.949$ on the stable branch beside it instead of oscillating. The matching equilibrium of the full seven-dimensional system is at $V^* = -42.812$ mV with $X_P^* = 0.9494$, and its largest eigenvalue real part is -2.72×10^{-4} ms^{-1}, thus it is stable. This is the only regime in which the fast subsystem doesn't produce anything apart from an equilibrium to settle on.

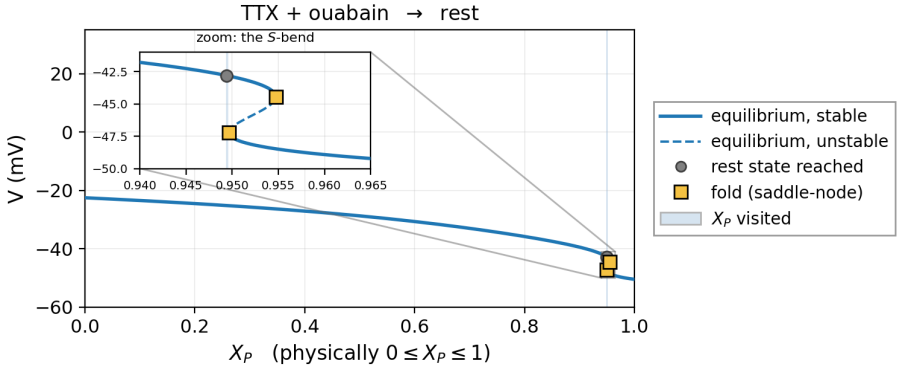


Figure 15: Fast-subsystem structure in X_P under combined blockade ($\alpha_I = \alpha_P = 0$). No spiking solution remains, and the equilibrium curve retains only a narrow S-bend where the zoom in showcases it between $X_P = 0.950$ and $X_P = 0.955$, too narrow to see on the full axis. The filled circle marks the rest state the full trajectory converges to, at $X_P = 0.949$.

5 Interpretation and Discussion

5.1 TTX Intervention

The model reproduces the central finding of Junge and Stephens [9] that TTX removes the fast action potentials but leaves the slow oscillatory wave (Figure 3b). The fast–slow analysis (Figure 9) shows how this arises. Spiking is lost because the stable spiking branches are destroyed, not because an equilibrium bifurcates. Once LP_2 lies within the range of X_P visited, the trajectory is reversed at the two folds and cycles between the stable branches with no action potentials, so the slow wave does not depend on spiking. This matches experimental observations,

since the slow wave relies on the persistent current g_T and the slow potassium feedback $g_P X_P$, neither of which TTX blocks. Recovery of spikes on washout corresponds to returning α_I to 1, although we did not simulate this.

Several limitations remain. The slow wave is attributed to X_P alone, but Y_A ($\tau_{Y_A} = 235\,\text{ms}$) is only a factor of 34 faster. The analysis lacks averaging techniques to determine the direction of flow along the 7D trajectory relative to the underlying bifurcation structures. Additionally, TTX is modelled as a fixed reduction of g_I, which ignores binding kinetics. Future work could address these in the model directly. A two-slow-variable (X_P, Y_A) analysis would test the timescale separation, and varying g_T, g_P and τ_{X_P} would show what sets the slow-wave period. Additionally, averaging methods could be implemented to map the invariant manifolds and vector field trajectories of the 7D system. Reducing the step size of the two-parameter continuation, and continuing each cusp and codimension-two point individually, would improve their locations and test whether any have been missed.

5.2 Ouabain Intervention

The model changes from bursting at $\alpha_P = 0.5$ to nearly regular firing during the final 20 s under complete pump blockade (Figure 10), which is consistent with experimental observations [9]. Fast–slow analysis shows that reducing the pump current shifts the resting state fold to $X_P \approx 0.960$, beyond the range covered by the complete trajectory, $X_P \in [0.783, 0.897]$ (Figure 12). Therefore, the trajectory continues spiking without reaching the resting branch, supporting the role for the pump current in organising bursting.

However, the bursting–beating transition has not been precisely located. At $\alpha_P = 0.2$, the firing pattern continues to change over the 60 s simulation, and its long-term behaviour is uncertain. The fold at $\alpha_P \approx 0.0941$ found by equilibrium continuation changes the number of equilibria, but its connection to the firing transition has not been established. The computed periodic-orbit branches locates outside the biological interval, with stability unverified away from Hopf points. The fast–slow analysis at intermediate inhibition shows how the trajectory and underlying structure change without establishing the transition boundary (Figure 13). Furthermore, ouabain is represented as a fixed scaling of the pump current, ignoring drug concentration dependence and inhibition kinetics. Since $I_{ext} = 0$ in the whole model, the experimental restoration of bursting by artificial hyper-polarisation [9] is not tested. Future research could use longer simulations and more precise parameter scans to locate the transition, supported by periodic orbit continuation and stability analysis. Adding a hyper-polarising current I_{ext} could test whether bursting is restored during pump blockade, as Plant and Kim's model predicts [7]. A concentration-dependent model of inhibition helps to improve comparison with experiments.

5.3 Combined Drug Intervention

The model reproduces the finding of Junge D [9] (Figure 3c) that under complete TTX, the slow wave is lost to a steady rest potential when ouabain removes the pump current, as Hypothesis 3 expected. With $\alpha_I = 0$ the pump acts as a control parameter for the slow wave. The slow wave amplitude shrinks to zero at the Hopf point H_{bio} ($\alpha_P \approx 0.27$), below which only a stable rest potential remains (Figure 14). The time traces and slow variable analysis agree with this threshold, although the traces only bracket it and the Hopf point was located by continuation. Since I_{ext} enters the voltage equation in the same way as I_{ep}, the model predicts that a sufficiently large hyper-polarising current would restore the slow wave, consistent with the experiments (Figure 3c), although we did not test this. The experiments also used calcium-free saline, whereas the model has no explicit calcium dynamics, so calcium-dependent effects are not represented.

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